

SIEGEL CENTER PROBLEM, KOLMOGOROV STYLE

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1. THE PROBLEM

For each $r > 0$, let $D_r := \{z \in \mathbb{C} : |z| \leq r\}$ and, given $R > 0$, let $T, T_* : D_R \rightarrow \mathbb{C}$ be defined by

$$\begin{aligned} T_*(z) &= e^{i\alpha} \\ T(z) &= T_*(z) + f(z), \end{aligned}$$

where f is holomorphic in a neighborhood of D_R , and $f(0) = f'(0) = 0$.

In addition we require that α is a Diophantine irrational, that is that there exists $p \in \mathbb{N}$ such that¹.

$$(1.1) \quad |e^{i\alpha n} - e^{i\alpha}| \geq Cn^{-p} \quad \forall n \in \mathbb{N}.$$

The question is if there exists $r > 0$ and an invertible function $H : D_r \rightarrow \mathbb{C}$, holomorphic in D_r , such that $H^{-1} \circ T \circ H = T_*$. That is if T is analytically conjugate to its linear part. In fact, we will show that such an H exists with the form $H(z) = z + h(z)$ with $h(0) = h'(0) = 0$.

Hence we are looking for h such that

$$(1.2) \quad e^{i\alpha}h(z) - h(e^{i\alpha}z) + f(z + h(z)) = 0.$$

To solve a functional equation it is usually necessary to specify the space of function in which we are looking for a solution.

2. SOME BANACH SPACES

Consider the function holomorphic in a neighborhood of some disk D_r . Clearly each such function can be written as $g(z) = \sum_{k=0}^{\infty} g_k z^k$. We can then define the two norms

$$(2.1) \quad \begin{aligned} \|g\|_r &:= \sup_{|z| \leq r} |g(z)| \\ |g|_r &:= \sum_{k=0}^{\infty} |g_k| r^k. \end{aligned}$$

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¹A natural question is: do such number exists? To see it consider the sets $I_{m,n} := [\frac{2\pi m}{n} - Cn^{-p}, \frac{2\pi m}{n} + Cn^{-p}]$. If $0 < m < n$, then $I_{m,n} \subset [0, 2\pi]$. Clearly if $\alpha \notin I_{m,n}$ for all $n \geq m \in \mathbb{N}$, then α satisfies the Diophantine condition. But $\sum_{n \geq m} |I_{m,n}| \leq C \sum_{n=1}^{\infty} n^{-p+1}$ which converges provided $p > 2$ and can be made arbitrarily small by choosing C small. Accordingly, almost all numbers are Diophantine for some $p > 2$. In fact, there exists also numbers that are Diophantine with $p = 2$, this are the ones that can be represented by *continuous fractions* with bounded entries. In particular, the quadratic irrationals have periodic continuous fractions hence they are Diophantine with $p = 2$. If you are unaware of these facts, look at the notes *Numeri razionali e reali*, sezione 3, and *Una digressione sul calendario*, sezioni 2.1, 2.3 at the web page <http://www.mat.uniroma2.it/~liverani/Inform06/didattica.html>

In this way we have the two normed spaces $\bar{\mathcal{B}}_r$ and $\mathcal{B}_r$ of the holomorphic functions g such that $\|g\|_r < \infty$ and $|g|_r < \infty$, respectively.

Problem 1. *Prove that $\bar{\mathcal{B}}_r$ and $\mathcal{B}_r$ are Banach spaces.*

Note that the two norms have the simple relation:

$$(2.2) \quad \|g\|_r \leq |g|_r.$$

The other direction is a bit less simple:

$$(2.3) \quad |g_n| \leq \left| \frac{1}{2\pi i} \int_{|z|=r} \frac{g(\xi)}{\xi^{n+1}} d\xi \right| \leq r^{-n} \|g\|_r$$

thus

$$(2.4) \quad |g|_{e^{-\tau}r} \leq \sum_{k=0}^{\infty} e^{-k\tau} \|g\|_r \leq 2\tau^{-1} \|g\|_r.$$

3. KOLMOGOROV'S IDEA

The basic idea is to solve the equation

$$(3.1) \quad L_f(h)(z) := e^{i\alpha}h(z) - h(e^{i\alpha}z) + f(z) = 0,$$

instead of equation (1.2), hoping to obtain a conjugation with a map with a smaller non linear part. Let us consider some $r \in (0, R)$ to be chosen later, then²

$$h_n = \frac{f_n}{e^{ian} - e^{i\alpha}}.$$

Remark 3.1. *In the following we will use C to designate a generic constant that depends only on α . Note that different occurrences of C may stand for different numerical values.*

Then³

$$(3.2) \quad |h|_{e^{-\tau}r} = \sum_{n=2}^{\infty} |h_n| r^n e^{-n\tau} \leq C \sum_{n=2}^{\infty} |f_n| r^n e^{-n\tau} n^p \leq C |f|_r \tau^{-p}.$$

We have thus a solution, albeit on a smaller disk. From now on, since we are interested in small losses in the radius, we will assume

$$(3.3) \quad \tau \in (0, 1].$$

The next step is to see to what T is conjugated by H , to do so we need to verify the existence of H^{-1} in some domain. Clearly, all is needed is that $H(z) = H(\zeta)$ implies $z = \zeta$ for each z, ζ in the given domain. This is like asking that $z - \zeta + h(z) - h(\zeta) = 0$ has the unique solution $z = \zeta$. Indeed, since

$$|z - \zeta| \leq \|h'\|_{e^{-\tau}r} |z - \zeta|$$

and, by (2.2) and (3.2),

$$(3.4) \quad \|h'\|_{e^{-\tau}r} \leq |h'|_{e^{-\tau}r} \leq C \sum_{n=2}^{\infty} |f_n| r^{n-1} e^{-n-1\tau} n^{p+1} \leq C |f|_r \tau^{-p-1} r^{-1},$$

²Note that here we use that $2\pi\alpha \notin \mathbb{Q}$, otherwise we would divide by zero!

³Here, instead, is the point where we use that not only $2\pi\alpha \notin \mathbb{Q}$, but it is a Diophantine irrational, that is it satisfies equation (1.1).

we have invertibility provided⁴

$$(3.5) \quad \tau > (Cr^{-1}|f|_r)^{\frac{1}{p+1}}.$$

We can finally set

$$H^{-1} \circ T \circ H(z) =: e^{i\alpha}z + f_1(z) =: T_1(z),$$

and verify if it is well posed and if f_1 is smaller than f as hoped.

To address such questions it is convenient to use the equivalent equation $T \circ H = H \circ T_1$, that is

$$\begin{aligned} f_1(z) &= e^{i\alpha}h(z) - h(e^{i\alpha}z + f_1(z)) + f(z + h(z)) \\ &= h(e^{i\alpha}z) - h(e^{i\alpha}z + f_1(z)) + f(z + h(z)) - f(z) =: K_f(f_1)(z). \end{aligned}$$

where we have used the fact that h satisfy (3.1). Clearly, the above equation is meaningful only if the compositions make sense. In other words, if we want the above equation to be well defined for, e.g., all $|z| \leq e^{-2\tau}r$, then we need that $e^{-2\tau}r + \|h\|_{e^{-2\tau}r} \leq e^{-\tau}r$, so that $f(z + h(z))$ is well defined and $e^{-2\tau}r + \|f_1\|_{e^{-2\tau}r} \leq e^{-\tau}r$, so that $h(e^{i\alpha}z + f_1(z))$ is well defined. The first inequality is ensured by (3.5), while the second tells us that we are interested in the action of the operator K_f on $B_\rho := \{g \in \bar{B}_{e^{-2\tau}r} : \|g\|_{e^{-2\tau}r} \leq \rho\}$, for $\rho < \tau r$.

If $\varphi \in B_\rho$,

$$\|K_f(\varphi)\|_{e^{-2\tau}r} \leq \|f'\|_{e^{-\tau}r} \|h\|_{e^{-2\tau}r} + |h'|_{e^{-\tau}r} \|\varphi\|_{e^{-2\tau}r}$$

To estimate the derivative we can use again Cauchy formula.

Problem 2. *Show that there exists $C > 0$ such that, for each $|z| \leq e^{-\tau}r$, holds*

$$(3.6) \quad |f'(z)| \leq \left| \frac{1}{2\pi i} \int_{|z|=r} \frac{f(\zeta)}{(z-\zeta)^2} d\zeta \right| \leq C|f|_r \tau^{-1}.$$

Using (3.6), (3.2), (3.4) we have

$$\|K_f(\varphi)\|_{e^{-2\tau}r} \leq C|f|_r^2 \tau^{-1-p} r^{-1} + C|f|_r \tau^{-1-p} r^{-1} \rho.$$

we have thus $K_f B_\rho \subset B_\rho$ provided condition (3.5) is satisfied with some appropriate constant C and

$$(3.7) \quad \tau r > \rho \geq C|f|_r^2 \tau^{-1-p} r^{-1}.$$

On the other hand, for all $\varphi, \phi \in B_\rho$,

$$\|K_f \varphi - K_f \phi\|_{e^{-2\tau}r} \leq \|h'\|_{e^{-\tau}r} \|\varphi - \phi\|_{e^{-2\tau}r} \leq C|f|_r - r \tau^{-p-1} r^{-1} \|\varphi - \phi\|_{e^{-2\tau}r}.$$

Thus, again if condition (3.5) is satisfied with some appropriate constant C , we have a contraction, therefore the fixed point equation $K_f f_1 = f_1$ has a unique solution in B_ρ . The last point is that we wish to have an estimate in term of the $|\cdot|$ norm and to do so we must use (2.4) and lose another bit of domain

$$(3.8) \quad |f_1|_{e^{-3\tau}r} \leq 2\tau^{-1} \|f_1\|_{e^{-2\tau}r} \leq 2\tau^{-1} \rho.$$

To summarize the situation, we have just proved the following.

⁴Note that, by definition, if $r \leq R/2$, we have $|f|_r \leq C_0 r^2$, thus $\tau \leq 1$ can always be achieved by choosing r sufficiently small.

Lemma 3.2. *There exists $R, C_1, C_2 > 0$ such that, for all $r < R$, $c \leq C_1$, $\tau \in (0, 1]$ and $f \in \mathcal{B}_r$ such that*

$$\begin{aligned} |f|_r &\leq c\tau^{p+1}r \\ C_2|f|_r^2\tau^{-1-p}r^{-1} &< r\tau \end{aligned}$$

one can solve $L_f(h) = 0$ and $K_f(f_1) = f_1$ so that $f_1 \in \mathcal{B}_{e^{-3\tau}r}$ and

$$|f_1|_{e^{-3\tau}r} \leq 2C_2|f|_r^2\tau^{-2-p}r^{-1}.$$

Problem 3. *Prove that, if $c \leq \min\left\{C_1, \frac{r}{4C_2}, C_2^{-\frac{1}{2}}\right\}$, then the condition $|f|_r \leq c\tau^{p+2}r$ implies $|f_1|_{e^{-3\tau}r} \leq \frac{1}{2}|f|_r$. (Hint: use Lemma 3.2.)*

4. THE ITERATION SCHEME

We are finally ready to set up an iteration scheme of which the previous section will constitute a generic step. Let $f_0 := f$, and let $r_0 < R$. Next, set $\tau_n := (n+1)^{-2}$, $r_{n+1} = e^{-3\tau_n}r_n$ and, for all $n \in \mathbb{N}_*$, given f_n satisfying the hypotheses of Lemma 3.2, call h_{n+1} the solution of $L_{f_n}(h) = 0$ and let f_{n+1} the the unique solution of $K_{f_n}(f_{n+1}) = f_{n+1}$.

To see that the above procedure defines an infinite sequence f_n , we must check

$$(4.1) \quad |f_n|_{r_n} \leq c\tau_n^{p+2}r_n$$

at each step. First of all notice that $r_n = e^{-3\sum_{k=1}^n k^{-2}}r > e^{-3}r_0 =: r_*$. Next, since $|f_0|_{r_0} \leq C_0r_0^2$, we have that f_0 satisfies the hypotheses of Lemma 3.2 provided $C_0r_0 \leq c$. Finally, let us assume that (4.1) is verified up to n , then

$$|f_{n+1}|_{r_{n+1}} \leq 2^{-n-1}|f_0|_{r_0} \leq 2^{-n-1}C_0r_0^2.$$

Then (4.1) is satisfied if $C_0r_0(n+2)^{2p+4} \leq c2^{n+1}$ which can be achieved by first choosing c and then r_0 small enough.

To conclude, set $T_n(z) := e^{i\alpha}z + f_n(z)$. By the results of the previous section holds

$$H_n^{-1} \circ T_n \circ H_n = T_{n+1}.$$

Hence, setting $\Psi_n := H_0 \circ H_1 \circ \dots \circ H_{n-1}$, holds $\Psi_n^{-1} \circ T_0 \circ \Psi_n = T_n$ and

$$\lim_{n \rightarrow \infty} T_n(z) = e^{i\alpha}$$

for each $|z| \leq r_*$. Moreover,

$$\Psi_n = \Psi_{n-1} + h_0 \circ \Psi_{n-1} = \Psi_0 + \sum_{k=0}^{n-1} h_k \circ \Psi_{n-k-1}$$

where we have set $\Psi_0(z) = z$. Hence

$$|\Psi_n - \Psi_0|_{r_*} \leq \sum_{k=0}^{n-1} |h_k|_{e^{-\tau_k}r_k} \leq C \sum_{k=0}^{n-1} |f_k|_{r_k} \tau_k^{-p} \leq C \sum_{k=0}^{n-1} 2^{-k-1}(k+1)^{2p} \leq C$$

This means that the series is norm convergent, hence it defines a function H holomorphic on the disk of radius r_* such that $H^{-1} \circ T \circ H = T_*$ that is, in such a disk the map is conjugated with its linear part.

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